Pathibhara Accounting System

User Manual

Complete Guide: Workflow, Features, Rules and Technical Design

Pathibhara Area Development Committee, Taplejung

Who this is for	Office head (Super Admin), accountant, auditor, and any developer who maintains the code
System version	Phase 1 + Income, commit `bda80d2`
Fiscal year set up	FY 2082/83 (2082.04.01 to 2083.03.end)
Source code	github.com/nabinbaral01/lekha

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1. What the system does: the one big idea

This is a cash-basis government accounting system that follows the Auditor General (FCGO) formats. It replaces the paper books the office keeps today.

> **The golden rule: enter each transaction once.** Every transaction is entered once, as a **journal voucher (Form 203)**. When the office head approves it, the system writes it into **every book automatically**: the bank-cash book, the budget statement, the tax register, the income report, the dashboard charts and the audit log. No book is typed separately, so the books can never disagree with each other.

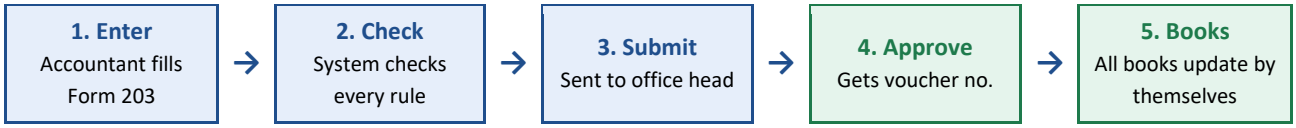
The four funds

Money is kept in four separate funds. Every voucher belongs to exactly one fund.

	Capital: assets, construction, purchases	**Recurrent: operations, services, programmes**
Internal: the committee's own revenue	Internal – Capital	Internal – Recurrent
Grant: money from government or other bodies	Grant – Capital	Grant – Recurrent

The four funds share **one real bank account**. The system tracks each fund's share of it and never lets the total bank balance go below zero.

How a transaction flows



Main features at a glance

Entry Form 203 voucher (easy payment mode or full debit/credit mode), income entry, tax deposit voucher, payee register, budget sub-heads.	**Control** Approval workflow, budget check, bank-balance check, period locks, date-order numbering, duplicate-bill warning, role permissions.
Books Bank-Cash Book (per fund and consolidated), Budget Statement, TDS pending list, income by source, dashboards with charts.	**Trust** Tamper-proof audit log (hash-chained and blocked from editing at database level), exact money (paisa, no rounding errors), Nepali/English, Nepali dates (BS).

2. Users and what each one can do

There are three roles. Every login belongs to one role, and the server checks the role on every action, so hiding a button is never the only protection.

Action	Super Admin (Office Head)	Accountant	Auditor
See all funds, vouchers, books, reports, charts	Yes	Yes	Yes
Print vouchers and books	Yes	Yes	Yes (logged)
Create and edit draft vouchers, income entries, payees	Yes	Yes	No
Submit a voucher for approval	Yes	Yes (own vouchers)	No
Cancel a draft or submitted voucher	Yes	Yes (own vouchers)	No
Approve or reject a voucher	**Yes (only this role)**	No	No
Approve spending above budget (with written reason)	Yes	No	No
Mark a voucher as paid	Yes	Yes	No
Settings: organisation, budget, heads, sub-heads, users, tax rates, opening balance, voucher numbering, month locks	Yes	No	No
Consolidated cash book (all four funds)	Yes	No	Yes
Audit log and tamper check	Yes	No	Yes

Demo logins

Username	Password	Role
admin	`Admin12345`	Super Admin
lekha	`Lekha12345`	Accountant
auditor	`Audit12345`	Auditor

> **Login safety.** After **5 wrong passwords** an account is locked for **30 minutes**. A session also ends by itself after **30 minutes without activity** (changeable in Settings). Change all demo passwords before entering real office data.

3. Starting, stopping and sharing the site

The project folder `C:\chetra 2` has three double-click files:

File	What it does
start.bat	Starts the website and its own database (one minimised window called "Pathibhara Accounts – do not close"). It waits until the site answers, then opens `http://localhost:3000` in the browser.
share.bat	Creates a free public link (`https://….trycloudflare.com`) so anyone on the internet can open the site. It prints the link and copies it to the clipboard. The link works only while this PC is on, connected, and the window stays open. A new link is made every time.
stop.bat	Stops the share link, the website and the database.

Ways to reach the site

Who	Address
On the same PC	`http://localhost:3000`
Other PCs in the office (same network)	`http://<this-PC-name>:3000`
Anyone on the internet	The link printed by share.bat
Permanent cloud link (optional)	Render.com using the ready-made `render.yaml`. It works even when the PC is off. The free plan sleeps after 15 minutes idle, and its free database expires after 30 days.

> Keep the office PC from sleeping (Windows Settings → System → Power → Sleep: Never) while it acts as the server.

4. Map of every screen

Navigation goes **Home → Internal/Grant → Capital / Recurrent / Income**, and then the tabs inside a fund.

Screen	Menu label	What it shows / does	Who
Home	Home	Summary tiles (shared bank balance, awaiting approval, TDS to deposit, days left in FY), spending-by-fund chart, and the Internal / Grant cards	All
Source page	Internal or Grant	Three cards: Capital expenditure, Recurrent expenditure, Income	All
Fund → dashboard	Dashboard	Budget, spent, remaining, status counts, money-flow chart	All
Fund → new voucher	Voucher Entry	Form 203 entry (easy payment or debit/credit mode)	Admin, Accountant
Fund → voucher list	Voucher List	Search, filter by status, open a voucher to print / submit / approve / reject / cancel / mark paid	All
Fund → cash book	Bank-Cash Book	Bank-cash book with running balances and monthly totals	All
Fund → tax	Tax (TDS)	Pending tax and the tax deposit voucher	All (create: Admin, Accountant)
Fund → budget statement	Budget Statement	Allocated / spent / saving per head	All
Fund → next phase	Budget Ledger · Advance · Reports	Placeholders for coming features	All
Income	Income	Income by head, new income entry, income voucher list	All (entry: Admin, Accountant)
Consolidated	All funds (consolidated)	Cash book of all four funds together: the true bank balance	Admin, Auditor
Audit	Audit Trail	Every action logged, with a tamper check	Admin, Auditor
Settings	Settings	Organisation · Budget sub-heads · Budget · Opening balances · Voucher numbering · Period locks · Users · Tax rates · Accounts / heads · Payees (details in chapter 5)	Admin
Top bar	—	Fiscal year · Nepali/English · Devanagari/Latin digits · light/dark theme · logout	All

5. First-time setup (Super Admin)

Do these once, in this order, before the accountant starts entering vouchers. Everything is under **Settings**.

#	Tab	What to enter	Why it matters
1	Organisation	Office name, government level, ministry, office code, PAN, address, the names and posts of the Accounts Head and Office Head	Printed on Form 203, the cash book and statements
2	Accounts / heads	Check the expense heads (code, activity no., name, Capital/Recurrent) and the income heads (code, Internal/Grant). Add the bank account name, branch and number.	48 expense heads and 11 income heads are pre-loaded with **provisional** codes. Correct them before first use; a head's code is frozen once a voucher uses it.
3	Budget sub-heads	Budget sub-head numbers and names from the red book (budget estimate)	Chosen on each voucher
4	Budget	Appropriation per head per fund. Later changes are added as: additional budget, reduction, or transfer between heads	Each change is a separate row with a note, so the history is kept. The budget of a head is the sum of its rows.
5	Opening balances	Opening bank and cash balance of each fund on 2082.04.01	The first row of the cash book. Cannot be changed once any voucher is approved.
6	Voucher numbering	If paper books already reached, say, voucher 159, set the next number to 160	Can be set only once, before the first approval
7	Users	Create real users, give roles, set strong passwords, deactivate demo users	Approval rights depend on role
8	Tax rates	Confirm TDS rates (have a Chartered Accountant check them)	A new rate is added with an effective date; old vouchers keep the old rate
9	Payees	Register regular suppliers with PAN (9 digits), VAT status and bank account	Needed for TDS, EFT and the tax register

6. Daily work: entering a payment voucher

Go to **Home → Internal or Grant → Capital or Recurrent → Voucher Entry**. There are two modes.

A. Simple payment (easy payment mode): used for almost everything

1. **Date** in BS, format `YYYY.MM.DD`. It must be inside the open fiscal year and in an unlocked month.
2. **Budget sub-head** and project name (optional).
3. **Expense head**: the list shows only heads of this fund's type (capital heads in Capital funds, recurrent heads in Recurrent funds) and the **remaining budget** next to each. The folio no. and activity no. fill in automatically from the head.
4. **Payee**: pick from the register or add a new one on the spot.
5. **Bill amount, bill number, bill date**. Tick *"Bill includes 13% VAT"* if the bill includes VAT. This is allowed only for VAT-registered payees.
6. **Tax type**: the system calculates TDS and the net payment instantly (see chapter 8). To change the TDS, tick "Override TDS" and write a reason.
7. Several payees can be added to one voucher.
8. **Payment method**: EFT (needs the payee's bank account) or cheque.
9. **Narration**: a short description is required.
10. **Preview** shows the exact Form 203 as it will print. Then **save as draft** or **submit**.

The system builds the accounting lines itself:

```
Dr  Expense head ..... bill amount (gross)
    Cr  Bank ............... net payment (bill − TDS)
    Cr  TDS payable ......... tax withheld
```

B. Debit/Credit (advanced mode)

For adjustments and unusual entries, the accountant writes each Dr/Cr line by hand, with the source columns (paying agency, payment method). Total debit must equal total credit, and the screen shows the difference live.

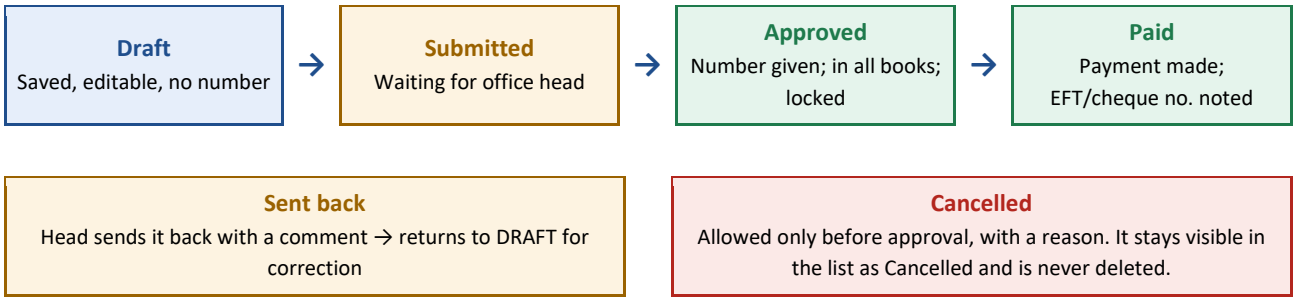
Warnings you may see while entering

- **Possible duplicate bill:** the same payee and bill number already exist, or the same payee and amount on the same day. You must tick "I have checked, this is not a duplicate" to continue.
- **Budget will be exceeded:** shown at submit. Only the Super Admin can approve it, and only with a written reason, which is recorded.

Printing Form 203

Open the voucher and press **Print**. The page is an exact copy of the paper Auditor General Form No. 203 on one A4 sheet, including the purpose and how-to-fill instructions text, the amount in words, and the signature blocks with names and BS dates. Each print is recorded in the audit log.

7. Voucher life: draft → submit → approve → paid



What happens at each step

Step	Who	What the system checks or does
Save draft	Accountant / Admin	All structural rules (dates, heads, PAN, Dr = Cr, VAT rules, duplicate warning). A draft can be edited any number of times.
Submit	The person who prepared it (or Admin)	The month must be open. The **budget check** runs: if any head would go over budget it stops, unless the Super Admin gives a reason.
Approve	**Super Admin only**	1. Only one approval at a time runs for the whole office, so numbers and balances never clash. 2. Date order: a voucher cannot be approved with a date earlier than the last approved voucher of that fund, so numbers stay in date order. 3. The budget check is repeated. 4. Bank check: the shared bank account (all four funds together) can never go negative. 5. For a tax deposit voucher, the amount must equal the TDS rows it clears. 6. Gives the next **voucher no.** for that fund and year: sequential, gap-free, never reused. 7. From now on the voucher appears in every book.
Reject	Super Admin	Comment required. The voucher goes back to DRAFT with the comment shown at the top.
Cancel	Preparer / Admin	Only DRAFT or SUBMITTED. Reason required. An approved voucher can never be cancelled or edited; a correction is made with a new adjustment (adjustment) voucher.
Mark paid	Accountant / Admin	Records that the bank payment was actually made, with the EFT/cheque reference.

> Two people editing the same voucher at once is detected: the second save is refused with "someone changed this voucher, reload" (version check), so no change is silently lost.

8. Tax deducted at source (TDS) and tax deposit

How the tax is calculated

1. If the bill includes 13% VAT, the taxable base = bill ÷ 1.13. Otherwise the base = bill.
2. TDS = base × rate of the chosen tax type. The rate used is the one **in force on the voucher date**.
3. Rounding follows the office setting. The default is **up to the next whole rupee**, which matches the client's books.
4. Net payment = bill − TDS.

All of this uses exact fractions, never decimals, so there is never a one-paisa error.

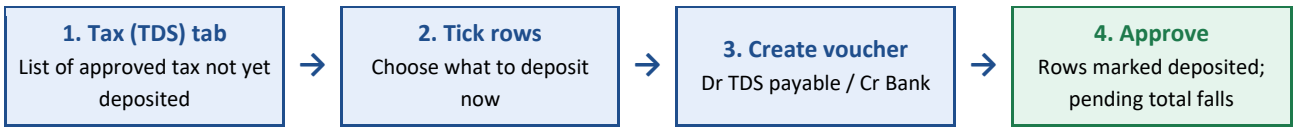
Tax type (default rates: verify with a CA)	Rate	Base
VAT-registered supplier/service (VAT_SUPPLIER)	1.5%	amount excluding VAT
Non-VAT bill (registered firm) (NON_VAT_FIRM)	1.5%	bill
Service/consultancy fee to an individual (INDIVIDUAL_SERVICE)	15%	bill
Salary/wages social security tax (SALARY_SST)	1%	bill
House / vehicle rent (RENT)	10%	bill
Exempt (EXEMPT)	0%	reason required

Worked example: the client's voucher 160

```
VAT bill (Consultancy services) ..............  79,100.00
Base = 79,100 ÷ 1.13 ....................  70,000.00
TDS  = 70,000 × 1.5% ....................   1,050.00
Net payment .............................  78,050.00

Lines:  Dr Consultancy services 79,100 | Cr Bank 78,050 | Cr TDS payable 1,050
Bank: 5,03,09,038.73 (opening) − 78,050 = 5,02,30,988.73  ✓ matches the paper cash book
```

Depositing the tax (tax deposit)



A row being deposited cannot be picked twice. If the deposit voucher is rejected or cancelled, the rows return to the pending list.

9. Income

Go to **Home → Internal or Grant → Income** (the green card). The page has three parts:

1. **Income by source:** every income head of that source with the amount received (approved), the amount awaiting approval, a share bar and the total.
2. **New income entry:** the entry form.
3. **Income vouchers:** every income voucher with its number, date, head, payer, receipt number, fund and status. Click the number to open or print it as Form 203.

Income heads loaded (provisional codes)

Internal (own revenue)	Grant (grants)
14211 Entry / ticket fee · 14213 Hotel, lodge & shop rent · 14212 Service fee · 14229 Parking fee · 14311 Donations & offerings · 14111 Bank interest · 14529 Miscellaneous income	13311 Federal government grant · 13312 Provincial government grant · 13313 Local government grant · 13314 Other agency / donor grant

The Super Admin can add more under **Settings → Accounts / heads**: choose type **REVENUE** and pick Internal or Grant.

Entering income

1. Date, and fund: which fund (Recurrent or Capital) the money is for.
2. Income head: only heads of this source are listed.
3. Amount; and where it went: Bank account or Cash.
4. Received from (person/body) is required; Receipt no. is optional.
5. Narration: if left empty, it becomes "Amount received for <head>".
6. **Submit** creates and submits it. The office head approves it like any voucher and it gets the next voucher no. of that fund.

```
Dr  Bank (or Cash) .......... amount
    Cr  Income head ......... amount
```

- Once approved, the bank balance in the cash book rises. The income is **not** counted as spending in the Budget Statement.
- The fund's chart shows an extra "Income" line.
- A Grant head cannot be used in an Internal fund, and the other way round. The system blocks it.
- On the printed Form 203 the payer and receipt number are added to the narration, and the payment table stays empty (nobody is paid).

10. Books and reports produced automatically

Every book below is **calculated** from approved vouchers each time it is opened, so it is never stored separately and never goes out of date. Each book has a date range filter (from/to, BS) and prints on A4.

Bank-Cash Book

Column	Filled from
Date, voucher no., description, cheque no.	Voucher date, number, narration, EFT/cheque reference
Cash: debit / credit / balance	Lines on the cash account, with running balance
Bank: debit / credit / balance	Lines on the bank account, with running balance
Budget expense	Net of expense-head lines
Advance: given / settled	Advance account lines
Miscellaneous: debit / credit	Everything else (TDS payable, income heads, deposits)

It starts with the **opening balance** row, and for each month shows **month total** and **balance carried forward**, then the yearly total and closing balance. The **consolidated** version merges all four funds and shows the true balance of the one shared bank account.

Budget Statement

For each head: **allocated** (sum of all budget rows) − **spent** (approved spending) = **saving**. Saving is always calculated, never typed.

Dashboards and charts

- **Home:** shared bank balance, vouchers awaiting approval, TDS to deposit, days left in the year. Each source card shows its budget, spending and %. The chart shows spending per fund.
- **Fund dashboard:** budget, spent, remaining, counts by status, rejected vouchers. The money-flow chart has lines for budget expense, bank payment, TDS and income. You can view it by month for the whole year, by week for one month, or as cumulative spending against the budget line. Every chart also has a table view.
- **Source page:** capital, recurrent and income cards with totals.

Acceptance check against the client's paper books

The client's figures were entered and every running balance matched to the paisa (opening 5,03,09,038.73; V160 → 5,02,30,988.73; V161 → 5,02,29,938.73; V162 → 4,97,83,480.73 …). These are kept as automatic tests (chapter 17).

11. Monthly and year-end routine

Every day

- Accountant enters and submits vouchers and income. The head approves or rejects them from the Voucher List (filter "Submitted"). After paying, mark vouchers Paid.

Every month

1. Check that no voucher of the month is still DRAFT or SUBMITTED.
2. Deposit the month's TDS from the Tax (TDS) tab.
3. Print the month's cash book page and compare the bank balance with the bank statement.
4. **Lock the month** (Settings → Period locks). After locking, nobody can create, submit or approve a voucher dated in that month. Only the Super Admin can unlock it, and both actions are logged.

Year end (Ashadh end)

1. Clear all drafts and pending vouchers. The year cannot be closed while any remain.
2. Print the yearly cash book, Budget Statement and audit log.
3. Create the next fiscal year and enter its opening balance (the closing balance of this year).
4. Close the old year.

12. Every safety rule and error message

All rules run on the server, whatever screen or tool is used. The messages appear in Nepali or English.

Rule	When it stops you
Debit = Credit	Totals differ on any voucher
Amounts positive	Zero or negative line amount (also enforced inside the database)
Budget	Spending would exceed the head's remaining budget; only the Super Admin may override, with a reason of at least 5 characters
Bank never negative	Approval would take the shared bank (or cash) below zero
Head type	Capital head in a Recurrent fund or recurrent head in a Capital fund
Income source	Grant income head in an Internal fund or the other way round
Date	Invalid BS date, a date outside an open fiscal year, or a date in a locked month
Date order	Approving a voucher dated before the fund's last approved voucher
PAN	Not exactly 9 digits
EFT	Payee has no bank account
VAT	VAT ticked for a payee not registered for VAT
TDS change / exemption	No reason given; changed TDS below 0 or above the bill
Duplicate	Same payee and bill no., or same payee and amount on the same day (needs confirmation)
Narration	Empty narration
Approved = final	Editing or cancelling an approved voucher
Head in use	Changing the code or type of a head or sub-head already used in a voucher (the name can still be corrected)
Opening balance	Changing it after any approval; an opening balance on an expense or income head
Database down	"Database is not running": restart with start.bat (the server also restarts its database by itself every 10 seconds)

13. Audit trail and security

- **Everything is logged:** logins (including failed ones), creating, editing, submitting, approving, rejecting, cancelling and printing vouchers, and every settings change. Each entry records who, when, from which computer, and the data before and after.
- **Nobody can change the log.** A database trigger blocks UPDATE, DELETE and TRUNCATE on the log table, even for the database administrator or a bug in the program.
- **Tamper check:** each entry stores a fingerprint (SHA-256 hash) of itself plus the previous entry, forming a chain. **Audit Trail → Verify** recomputes the chain and shows either "intact" or the exact entry that was altered.
- **Passwords** are stored only as bcrypt hashes. Login tokens are stored as SHA-256 hashes, so a stolen database cannot be used to log in.
- **Auditor** is read-only everywhere; the server refuses any change from that role.
- **Money** is stored as whole paisa (BIGINT). There are no floating-point numbers anywhere, so totals are always exact.

14. Technical design (for developers)

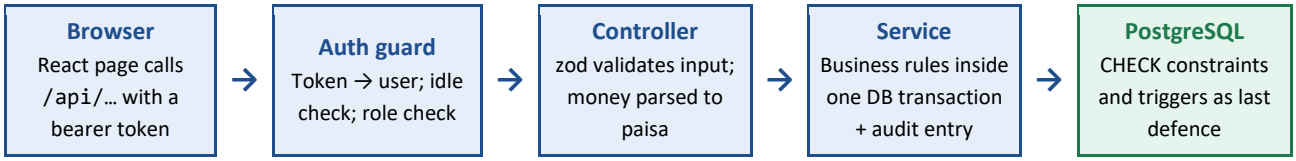
Technology

Layer	Technology
Browser app	React 18 + TypeScript, Vite, Tailwind CSS, TanStack Query, react-router, react-i18next (ne/en), offline fonts
Server	NestJS 11 (Node.js 22+), zod validation, bcryptjs; it also serves the built browser app
Database	PostgreSQL (UTF-8) through Prisma 6. In single-PC mode (EMBEDDED_PG=1) the server starts its own PostgreSQL on port 5544.
Shared core	`@pa/core`: money, TDS, BS calendar, amount in words, voucher rules. The same code runs in the browser and on the server, so the preview always equals what is saved.

Folder layout

```
C:\chetra 2
├── start.bat · stop.bat · share.bat     double-click helpers (chapter 3)
├── render.yaml                          one-click cloud deployment
├── packages/core/src                    @pa/core — shared accounting core
│   ├── money.ts      parse/format paisa, Nepali digit grouping (5,03,09,038.73), Devanagari digits
│   ├── tax.ts        computeTds(): exact-fraction TDS with rounding modes
│   ├── bsdate.ts     BS ⇄ AD conversion, today in Nepal time, fiscal-year helpers
│   ├── words.ts      amount in words (Nepali / English)
│   └── voucher.ts    line validation, payment/tax-deposit line builders, cash-
book columns, PAN check
├── apps/server
│   ├── prisma/schema.prisma            database model (chapter 15)
│   ├── prisma/migrations/              init · integrity guards · head activity & paid · income
│   ├── prisma/seed.ts, seed-demo.ts    empty office / demo data
│   ├── scripts/add-heads.ts            add new standard heads to an existing office
│   ├── test/acceptance.e2e.test.ts     31 end-to-end tests with client figures
│   └── src
│       ├── main.ts, bootstrap.ts       start-up, .env, embedded DB watchdog, serve web app
│       ├── auth/                       login, sessions, idle timeout, lockout, role guard
│       ├── audit/audit.service.ts      hash-chained audit log
│       ├── master/                     settings: org, years, locks, heads, sub-
heads, payees, tax, budget, opening, users, audit
│       ├── vouchers/vouchers.service.ts  compose + validate, submit, approve, reject, cancel, paid,
 tax deposit
│       ├── vouchers/ledger.ts          balance/budget queries from approved lines
│       ├── reports/reports.service.ts  cash book, budget statement, trend, income, fund summaries
│       ├── common/                     errors, DB-down handling, zod helpers, BigInt→JSON
│       └── seed/seed.ts                48 expense heads, 11 income heads, tax rules, 4 funds
└── apps/web/src
    ├── App.tsx                         routes (chapter 4)
    ├── pages/  Home · VoucherEntry · Vouchers · Income · Reports · Admin
    ├── components/  Form203 (exact paper form) · BsDateInput · PayeeForm · TrendChart · Layout · ui
    ├── i18n/ne.ts, en.ts               every label and error message
    └── lib/  api · session · prefs · types
```

How a request travels



Key design decisions

- **Books are derived, not stored.** `ledger.ts` sums approved voucher lines on demand. Nothing can drift out of sync.
- **One compose() function** builds the lines for preview, create and update, so what is previewed is exactly what is saved.
- **Money is BIGINT paisa**, sent over JSON as exact decimal strings ("78050.00").
- **Approvals are serialised** with a PostgreSQL advisory lock per office, and the voucher counter row is locked (`FOR UPDATE`), which gives gap-free numbers even with many users.
- **Optimistic locking:** each voucher has a version number, and stale edits are refused.
- **BS dates** are stored both as text (`2082.04.01`) and as an AD date for sorting and ranges.

15. Database tables

Table	Holds
Organization	Office details, signatories, VAT rate, TDS rounding, session timeout
User, Session	Logins, roles, lockout counters; active sessions (hashed tokens)
FiscalYear, PeriodLock	Fiscal years (BS and AD start/end, closed flag); locked months
Fund	The four funds (source × nature)
LedgerAccount	Chart of accounts. Kind = EXPENSE (with nature), REVENUE (with source), BANK, CASH, TDS_PAYABLE, ADVANCE, DEPOSIT, OTHER. Also holds code, activity no., folio no. and bank details.
BudgetSubHead	Budget sub-head numbers
BudgetEntry	One row per budget change (appropriation, addition, reduction, transfer pair)
OpeningBalance	Opening balance per fund and account
Payee	Payee register (PAN, VAT, bank)
TaxRule	TDS rate history with effective dates
VoucherCounter	Next voucher no. per fund and year
Voucher	Header: fund, year, number, type, mode (PAYMENT / MANUAL / TAX_DEPOSIT / RECEIPT), status, dates, narration, total, who prepared/submitted/approved/cancelled/paid and when, version
VoucherLine	Dr/Cr lines: account, kind, amount, activity, component, source columns
VoucherPayee	Payment table rows: payee, PAN, bank, bill, gross, TDS, net, link to the tax deposit voucher (for income: payer and receipt no.)
Attachment	Files attached to vouchers (bills, quotations)
AuditLog	Append-only, hash-chained history of every action

Database-level guards: amounts > 0; gross − TDS = net; an approved voucher must have a number; tax rate 0–100%; month 1–12; audit log immutable.

16. API reference

All paths start with `/api` and need `Authorization: Bearer <token>`, except login. Write actions are refused for the Auditor role.

Method and path	Purpose	Role
`POST /auth/login · POST /auth/logout · GET /auth/me`	Sign in / out, current user	all
`PATCH /auth/me/language · /auth/me/password`	Own language and password	all
`GET /meta`	Office, funds, years, today (BS), sub-heads, tax rules	all
`PATCH /organization`	Office settings	Admin
`POST /fiscal-years · POST /fiscal-years/:id/close`	New year / close year	Admin
`GET, POST /fiscal-years/:id/locks`	Month locks	Admin (GET all)
`GET /accounts · POST /accounts · PATCH /accounts/:id`	Chart of accounts (expense and income heads, bank…)	Admin (GET all)
`POST /sub-heads · PATCH /sub-heads/:id`	Budget sub-heads	Admin
`GET, POST /payees · PATCH /payees/:id`	Payee register	Admin, Accountant
`GET, POST /tax-rules`	TDS rates	Admin (GET all)
`GET, POST /budget-entries`	Budget rows	Admin (GET all)
`GET, POST /opening-balances`	Opening balance	Admin (GET all)
`GET, POST /users · PATCH /users/:id`	Users	Admin
`GET /audit · GET /audit/verify · POST /audit/print`	Audit log, tamper check, log a book print	Admin, Auditor
`GET /funds/:fundId/vouchers`	Voucher list (status, search, pages)	all
`POST /funds/:fundId/vouchers/preview`	Exact Form 203 preview	Admin, Accountant
`POST /funds/:fundId/vouchers`	Create draft (mode PAYMENT, MANUAL or RECEIPT)	Admin, Accountant
`GET /funds/:fundId/budget-position`	Budget, spent, pending, remaining per head	all
`GET /funds/:fundId/tds/pending · POST /funds/:fundId/tax-deposit`	Pending TDS; create tax deposit voucher	all / Admin, Accountant
`POST /funds/:fundId/voucher-start-number`	Starting voucher no. (once)	Admin
`GET /vouchers/:id · PUT /vouchers/:id`	Read / edit draft	all / Admin, Accountant
`POST /vouchers/:id/submit · /cancel · /paid`	Workflow	Admin, Accountant
`POST /vouchers/:id/approve · /reject`	Workflow	Admin
`POST /vouchers/:id/printed`	Record a print	all

`GET /reports/cash-book/:fundId · GET /reports/cash-book`	Cash book per fund / consolidated	all / Admin, Auditor
`GET /reports/budget-statement/:fundId`	Budget Statement	all
`GET /reports/income/:source`	Income by head + income vouchers	all
`GET /reports/trend · GET /reports/fund-summaries`	Chart data; dashboard cards	all

17. Testing

Run `npm test` from the project folder. Each run creates a fresh database, so tests never touch real data.

Suite	Count	Covers
Core unit tests	21	Money parsing and formatting, TDS rounding, BS calendar, amount in words, voucher rules
Client figures (§12)	12	Opening balance, V160–V183 TDS and net, running bank balances, budget savings, charts equal books, cash book reconciles to the paisa
No-mistake rules (§8)	10	Dr ≠ Cr, overspend + override, bank overdraft, head type, dates, locked month, duplicates, VAT/TDS reasons, PAN, date order
Roles and audit (§3)	7	Auditor read-only, accountant limits, approved is final, lockout, immutable log and intact chain, sub-heads, reject flow
Income	2	Dr bank / Cr income, source mismatch blocked, pending/received totals, bank rises, not counted as expense; only Admin adds income heads

Last run: **all 31 server tests and 21 core tests passed.**

18. Troubleshooting

Problem	Fix
"This site can't be reached" at localhost:3000	Double-click start.bat and wait for the browser to open.
"Database is not running" on login	Run stop.bat, then start.bat.
Public link does not open	The PC slept or share.bat was closed. Run start.bat, then share.bat, and send the **new** link.
"Account locked"	5 wrong passwords. Wait 30 minutes, or the Super Admin unlocks it in Settings → Users.
Logged out by itself	30 minutes without activity (session timeout). Log in again.
Cannot approve: date order	A later-dated voucher of this fund is already approved. Correct the date, or approve vouchers in date order.
Cannot approve: budget	Add budget (additional budget / transfer), or approve as Super Admin with a written reason.
Cannot approve: insufficient balance	The shared bank does not have enough money. Record the income or grant first.
Cannot enter a date	The month is locked, or the date is outside the fiscal year.
A head is missing from the list	It belongs to the other type (capital/recurrent, internal/grant) or is inactive. Check Settings → Accounts / heads.

19. What is still to be built, and what must be verified

Done

Schema, roles, login, audit trail · four-fund navigation · Form 203 entry with TDS engine and approval workflow · bank-cash book, budget statement and printing · tax deposit · dashboards and charts · bilingual interface · income (Income) · all acceptance tests.

Next phases

Feature	What it adds
Backup / restore	One-click backup of all data to a file and restore. **Recommended before go-live.**
Advance	Giving and settling advances; advance register with ageing
Budget Ledger (Budget ledger)	Page for each head with every voucher and its running balance
TDS register and IRD monthly report	Tax withheld per payee and PAN, ready for filing
Excel export	Cash book, Budget Statement and income report as .xlsx
Bank reconciliation, trial balance, other reports	Month-end checks
Licensing	Licence key per installation

Must be verified before go-live

1. Expense-head codes and activity no. (48 heads) against the office's approved budget book.

2. Income-head revenue codes (11 heads) against the office books.

3. All TDS rates, by a Chartered Accountant against the current Finance Act.

4. Client book findings: the TDS on paper vouchers V165 (8,884) and V185 (456) matches no standard rate. The Budget Statement shows less spending than the cash book for consultancy, festivals, service contracts, meeting allowance and furniture. Capital heads appear in the Recurrent statement. The paper books shared are FY 2081/82.

5. Replace all demo passwords and deactivate the demo users.

End of guide.